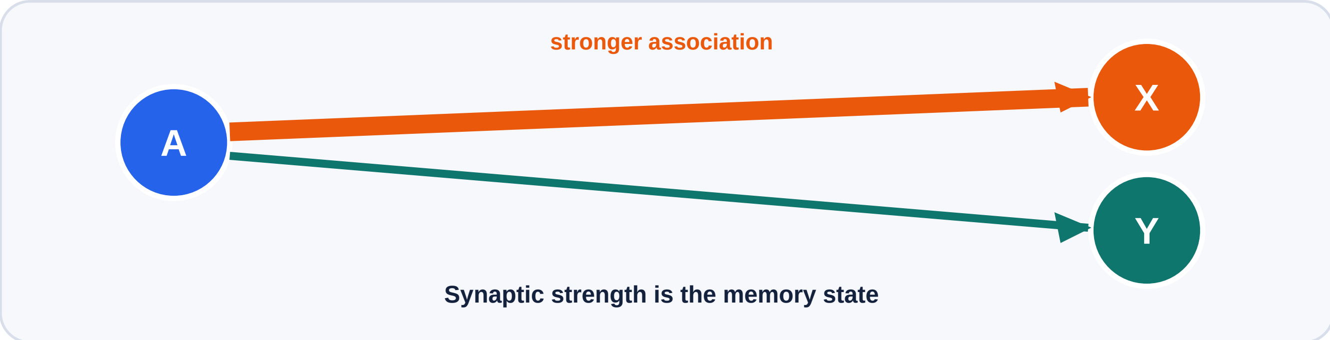
Synaptic Plasticity as Short Term Memory

How a changing synaptic state can learn, forget, and adapt



Central claim
Recent neural activity can temporarily change synaptic strength, allowing a small network to store short term associations. Decay and competing writes can make those memories interfere.

1. Where does the memory live?

In the interactive artifact, memory lives in the **current strength of connections**. Start with an empty 4 × 4 synaptic matrix. Write A → X and the A to X connection becomes stronger. Query A and the strongest outgoing association is recalled.

This makes an otherwise hidden internal state visible. The matrix gives exact values, while the live network view uses line thickness and brightness to show relative strength.

2. The learning rule

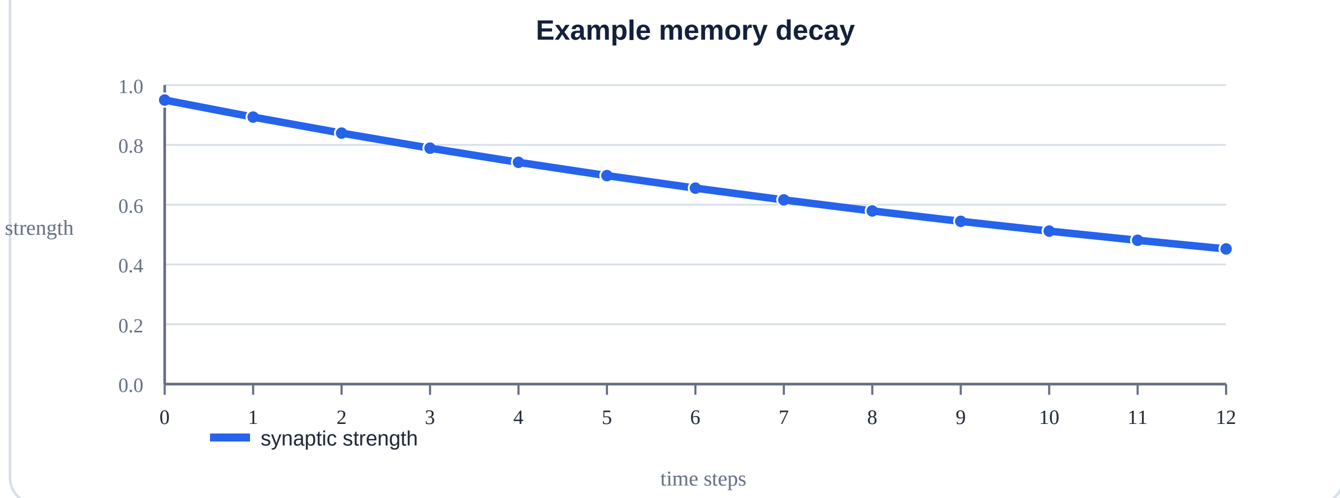
$$w(t+1) = \text{retention} \times w(t) + \eta \times \text{pre}(t) \times \text{post}(t)$$

w(t)	current synaptic strength	**retention**	how much of the old state survives
η	strength of a new write	**pre × post**	coincident activity that triggers the simplified write

This is an **educational toy rule**. It is not presented as the complete BDH equation or as a biologically complete simulation.

3. Learning, forgetting, and decay

A write strengthens a selected connection. When no new write is added, retention carries part of the old state forward, so the stored association gradually fades.

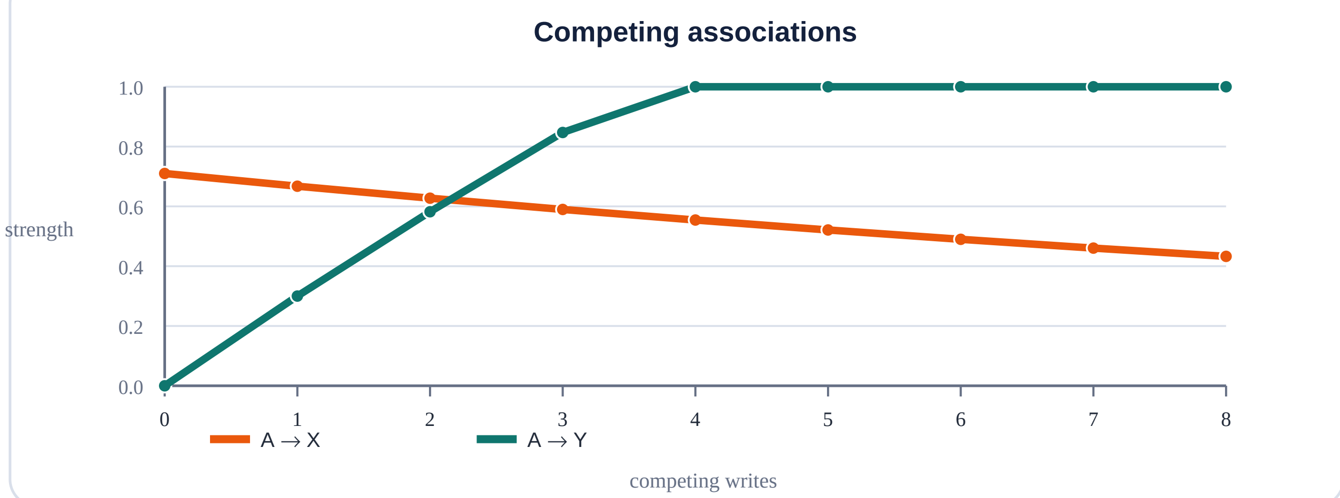


The x axis is numbered **0 to 12**. The curve is generated by the toy model and should not be interpreted as a biological time constant.

High retention	**Low retention**
More of the previous state survives.	The previous state fades faster.

4. Interference: when memories compete

Teach A → X first. Then repeatedly teach A → Y. Both memories use the same cue, so the new writes strengthen A → Y while the old A → X association continues to be affected by retention.

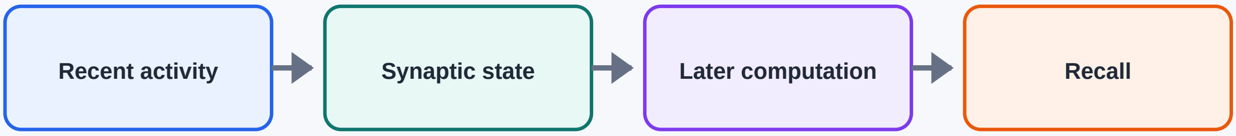


The x axis is numbered **0 to 8**. In the default toy run, A → Y becomes stronger than A → X after **three post switch writes**. That is a result of the toy rule and chosen parameters.

5. Plastic memory versus frozen memory

The comparison experiment first teaches A → X and then switches the environment to A → Y. Plastic memory keeps updating its synaptic state. Frozen memory keeps the old association.

Plastic Can adapt because its internal state continues to change.	**Frozen** Keeps the old association and cannot learn the switch.



Conceptual bridge to synaptic memory in BDH

With the default parameters, the toy model takes three post switch writes before A → Y overtakes A → X. This is a property of the displayed equation and parameters, not a biological measurement.

6. The BDH connection

BDH provides the broader motivation for treating memory as a changing synaptic state. The DataForge problem statement describes BDH as a brain inspired Post Transformer architecture where reasoning and memory share one computational fabric, with attention reformulated as synaptic memory.

Kosowski et al. (2025), the primary BDH paper, describes working memory during inference using synaptic plasticity and Hebbian learning with spiking neurons and reports strengthening of individual synapses when specific concepts are processed.

What our artifact demonstrates A small, transparent toy model in which activity changes a synaptic state and that state affects recall.	**What it does not claim** It does not reproduce the full BDH architecture, neuron dynamics, training procedure, scale, or benchmark results.

7. Six step learner journey

1 · Predict	What happens if retention is reduced?
2 · Write	Teach A → X and observe the matrix.
3 · Query	See which association is strongest.
4 · Decay	Advance time without a write.
5 · Compete	Teach A → Y and watch interference.
6 · Measure	Compare plastic and frozen memory.

8. Research context

Recent computational research studies plastic synaptic states as mechanisms for learning, forgetting, temporal structure, and adaptation. These sources ground the explanation; they do not validate every numerical behavior of the toy artifact.

Kosowski et al. (2025)
The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. Primary BDH source.

Rodriguez, Guo & Moraitis (2022)
Short-Term Plasticity Neurons Learning to Learn and Forget. ICML 2022, PMLR 162.

Brito & Gerstner (2024)
Learning what matters: Synaptic plasticity with invariance to second-order input correlations. PLOS Computational Biology.

Koprinkova-Hristova et al. (2023)
Learning cortical hierarchies with temporal Hebbian updates. Frontiers in Computational Neuroscience.

9. Learning outcomes

Explain	How a changing synaptic state can serve as temporary memory.
Observe	How repeated writes strengthen an association and retention produces decay.
Analyze	How competing writes create interference and adaptation.
Connect	How the toy mechanism relates to the synaptic memory idea in BDH.
Distinguish	Toy model behavior from published biological and BDH evidence.

10. Limitations and honest interpretation

The network is tiny and the update rule is simplified. It does not simulate realistic neuronal dynamics, biochemical mechanisms, or biological timescales. The adaptation delay is a property of the toy rule and selected parameters. The artifact does not report BDH benchmark performance.

11. Reproducibility and accessibility

The artifact uses HTML, CSS, and vanilla JavaScript. It has no external runtime JavaScript library, external font dependency, pretrained weights, or external dataset. The interface includes keyboard accessible controls, visible focus states, responsive layouts, readable labels, and reduced motion support.

12. References

[1] Kosowski, K. et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain.* arXiv:2509.26507.

[2] Rodriguez, M. G., Guo, Z., & Moraitis, T. (2022). *Short-Term Plasticity Neurons Learning to Learn and Forget.* ICML 2022, PMLR 162.

[3] Brito, A. & Gerstner, W. (2024). *Learning what matters: Synaptic plasticity with invariance to second-order input correlations.* PLOS Computational Biology.

[4] Koprinkova-Hristova, P. et al. (2023). *Learning cortical hierarchies with temporal Hebbian updates.* Frontiers in Computational Neuroscience.

Final takeaway
A synapse can be treated as a small piece of changing state. Recent activity can write into that state, retention can let it fade, and competing activity can change what is remembered. The interactive artifact lets the learner test that idea directly.